PROGRESS REPORT · AI & COMPUTER VISION

What Makes AI-Based Sparse-View 3D Gaussian Splatting Work?

Separating the roles of predicted camera poses, predicted 3D points and pose refinement

Team	Supervisor	Report date	Project status
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1. Summary

We are studying why a modern AI-based method (InstantSplat) can build a 3D scene from only 3 to 9 photos. We will switch its parts on and off, one at a time, to see which part creates the quality. In the first block we built and checked the full experiment setup.

Key results so far

Setup works: the full pipeline runs on free Google Colab and is shared with the team (Drive, GitHub, live tracker).

Milestone M1 passed: our baseline matches the published numbers within 0.6 dB (PSNR 14.70 / 16.39 / 17.19 vs 14.10 / 15.83 / 17.16 at 3 / 6 / 9 photos).

Two hidden problems found and fixed in the original code, so that all later comparisons are fair.

Main open point: computing time. All 96 planned runs need about 22.5 GPU-hours on the free GPU.

2. The project in plain words

What is 3D Gaussian Splatting?

3D Gaussian Splatting (3DGS) turns photos of a scene into a 3D model that can be viewed from new angles. It needs the camera pose of every photo: where the camera stood and where it pointed. With 100+ photos the classic tool (COLMAP) computes these poses well. With only 3 to 9 photos it often fails.

How AI helps

New AI models (here MASt3R) predict the camera poses and a 3D point cloud directly from a few photos. InstantSplat combines three parts: (1) AI-predicted poses, (2) AI-predicted 3D points, and (3) pose refinement, which corrects the poses during training. Nobody has tested which of these three parts matters most when only 3 to 9 photos are available.

Research question

How much of the quality comes from the AI's predicted poses, its predicted 3D points, and pose refinement, and how does this change from 3 to 9 photos? In particular: do the dense-view findings of VGGT-X still hold with only 3 to 9 photos?

Method: change one part at a time

Config	Camera poses	3D points	Refinement	What it tells us
C1	Reference (from the dataset)	AI (MASt3R)	Off	Best case for poses
C2	AI (MASt3R)	AI (MASt3R)	Off	Pure AI prediction
C3	AI (MASt3R)	AI (MASt3R)	On	Full InstantSplat (baseline)
C4	AI (MASt3R)	Random	On	Value of the AI points

Comparisons: C1 vs C2 = effect of poses · C3 vs C4 = effect of 3D points · C2 vs C3 = effect of refinement. Data: Mip-NeRF 360, 8 real 360° scenes, 3 / 6 / 9 training photos. Plan: 4 configs × 3 photo counts × 8 scenes = 96 runs. A fifth configuration with a newer AI model (VGGT) is optional.

3. What we have done

Date	Work	Outcome
3–6 Nov	Related work: Google Scholar search and the research tools Elicit, Consensus and Perplexity; about 30 papers read; research gap defined (Farida)	Our first idea was already published, so we sharpened it. Gap found: no study separates the three parts at 3–9 photos
3–6 Nov	Proposal, presentation and day-by-day plan for two people	Detailed proposal (PDF), slides and tracker; 4-block relay plan to 25 Nov
9 Nov	Set up InstantSplat on Google Colab (free T4 GPU)	Works after 4 compatibility fixes; setup cached on Drive so it restarts in minutes
9 Nov	Dataset and test rule	8 scenes ready; same rule as the Free360 paper: every 8th photo = test photo
9 Nov	Baseline: full InstantSplat (C3) on 2 scenes × 3 / 6 / 9 photos = 6 runs	Matches the published numbers (Section 4) → milestone M1 passed
9 Nov	Quality control of the original code	Two problems found and fixed (Section 5)
9 Nov	Team sharing	Shared Drive folder, private GitHub repo, live Google Sheet tracker, project log

4. First results

Before changing anything, we checked that our setup gives the same numbers as a published paper (Free360, which also tested InstantSplat). PSNR measures how close the rendered test images are to the real photos (higher is better).

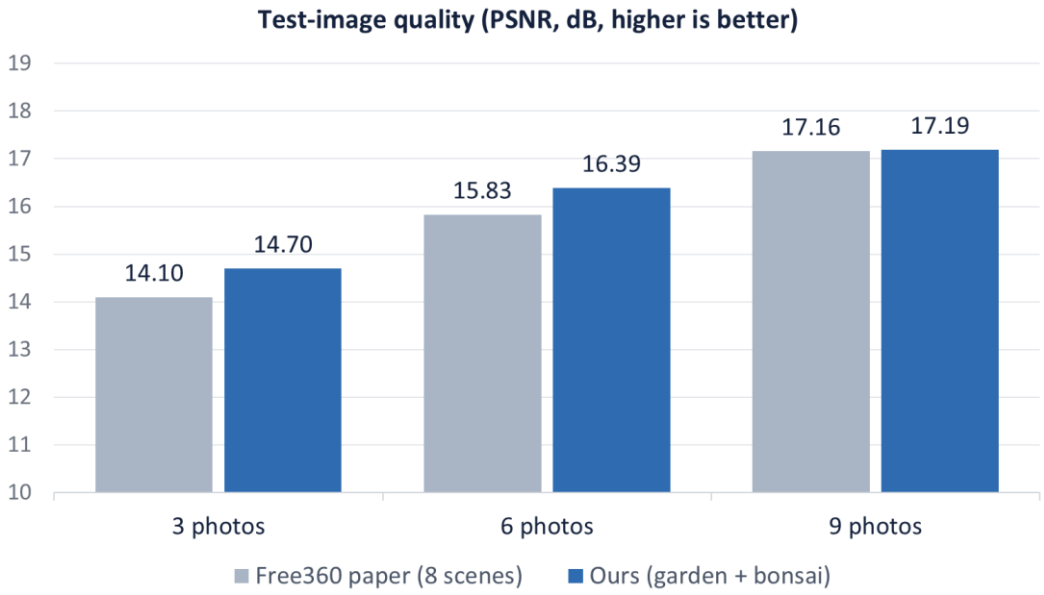


Figure 1. Test-image quality: our baseline vs the published Free360 numbers.

Photos	Our PSNR ↑	Free360 PSNR ↑	Difference	Our SSIM ↑ / LPIPS ↓	Free360 SSIM ↑ / LPIPS ↓
3	14.70	14.10	+0.60	0.315 / 0.568	0.296 / 0.529
6	16.39	15.83	+0.56	0.367 / 0.450	0.351 / 0.480
9	17.19	17.16	+0.03	0.389 / 0.413	0.402 / 0.443

Per scene (time and GPU memory per run on the free T4 GPU):

Scene	Photos	PSNR ↑	SSIM ↑	LPIPS ↓	Time (min)	GPU memory (GB)
garden	3	14.65	0.273	0.555	8.9	3.2
garden	6	17.58	0.345	0.396	12.4	3.7
garden	9	18.04	0.344	0.377	16.8	4.8
bonsai	3	14.75	0.358	0.581	9.9	3.3
bonsai	6	15.20	0.390	0.503	15.4	3.7
bonsai	9	16.34	0.434	0.449	21.1	4.9

What this means

Our setup is correct: every photo count is within 0.6 dB of the paper (our target was 1.5 dB), so we can trust the next experiments.

Quality improves with more photos, as expected. The outdoor scene (garden) gains most from 3 to 6 photos.

Caution: Free360 averages 8 scenes, we used 2, so small differences are normal.

5. Problems found and fixed

Problem	Fix	Effect
Before measuring, InstantSplat guesses where each test camera is. In 360° scenes this first guess was 80–90° wrong.	Start each test camera from its reference position, aligned to the AI's cameras. The same rule is used for every configuration.	garden PSNR rose from 14.23 / 14.52 / 14.76 to 14.65 / 17.58 / 18.04; the benefit of more photos is now visible
InstantSplat's camera-error code mixes two camera formats and gave impossible values (up to 145°).	We wrote our own script (pose_error.py) and checked it with automatic tests.	Valid camera-error numbers from the next runs (re-run planned)

Why this matters: both problems would have made the comparison between configurations unfair or misleading.

6. Computing time

- **14 minutes** per run on average on the free T4 GPU (9 to 21 minutes; more photos take longer).
- **65%** of each run is spent fine-tuning the test cameras before measuring. The method requires this step.
- **22.5 GPU-hours** for all 96 runs, about 11 hours for each of us.
- **Memory is not a problem:** at most 4.9 GB of the 15 GB GPU.

Options (decision before 16 Nov): split the runs over our two free Colab accounts as planned; buy Colab compute units for a guaranteed GPU; use the RTX 5090 workstation as back-up; or, as a last resort, reduce to 4 scenes (48 runs). All final runs will use one GPU type.

7. Status and next steps

Still open in Block 1 (by 11 Nov)

Task	Owner	Status
Re-run the baseline with the updated notebook (valid camera errors, dataset copy on Drive)	Farida	In progress
Mona runs the notebook once on her own account	Mona	Not started
Overleaf + Zotero set-up; get the DPR-Splat paper	Farida	In progress
15-minute hand-off call (Block 1 → Block 2)	Both	In progress

Plan to the deadline

Dates	Block (lead)	Goal
9–11 Nov	Block 1 (Farida)	Setup + baseline — main result done
12–15 Nov	Block 2 (Mona)	Add the switches C1, C2, C4; test each on 1 scene; 24 pilot runs on 2 scenes
16–18 Nov	Block 3 (both)	All 96 runs; results tables and figures (milestone M2)
19–23 Nov	Block 4 (Mona)	Analysis and paper draft; Farida checks every number
24–25 Nov	Final (both)	Send to Dr. Wedad Sallam; apply feedback; deliver

8. Questions for the supervisor

1. Is an analysis / ablation study acceptable, or is a new method required?
2. What is the target output: course report, thesis chapter or workshop paper?
3. Compute for the 96 runs: free Colab, paid Colab units or the RTX 5090 workstation?

9. Where to find everything

- **GitHub (private):** github.com/Farida-gaber44/sparse-view-3dgs-ablation — notebook, scripts, project log (LOG.md), slides.
- **Live tracker (Google Sheet):** linked in the GitHub README; both of us update the status there.
- **Shared Google Drive folder 3dgs-project:** setup files, dataset copy and all run results.
- **Colab notebook:** notebooks/01_block1_setup_baseline.ipynb — press Run all; finished runs are skipped.

Glossary

Term	Meaning
Camera pose	Where a camera stood and where it pointed when the photo was taken
Sparse-view	Building a 3D scene from only a few photos (here 3, 6 or 9)
Reference poses	Very accurate poses computed by COLMAP from all 100–300 photos of a scene (part of the dataset)
Pose refinement	Correcting the camera poses during 3D training
Ablation study	Turning parts of a method on or off, one at a time, to measure what each part contributes
PSNR / SSIM / LPIPS	Image-quality scores comparing rendered and real test photos (higher, higher, lower is better)
GPU / T4	Graphics card that runs the training; the T4 is the free GPU on Google Colab